

EECS 119
Project 4 : 4 Bit Up-Down
Jack Lu
12/4/25

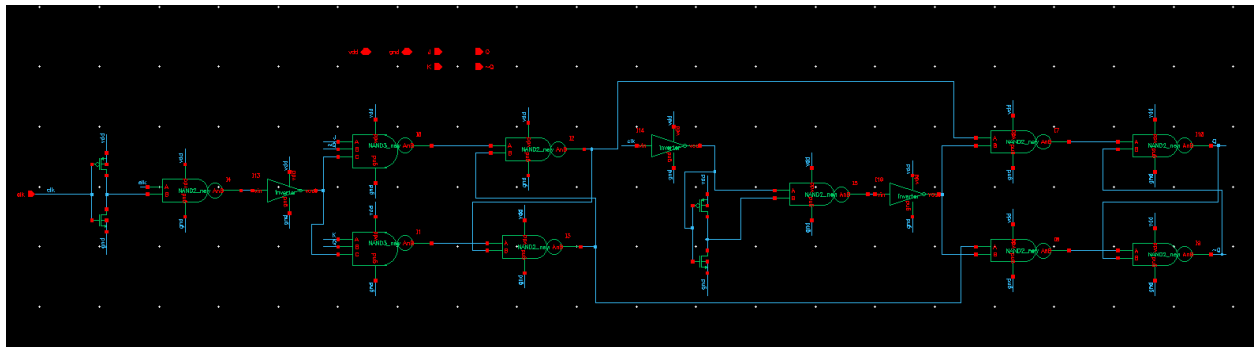
Introduction

For this project, we are to build and make a 4 bit up - down counter with JK Flip Flops by cascaded the individual flip flops, additionally, we are require to test the schematic and layout and make sure it pass with no error, and get the DRC and LVS match on layout to ensure simulation results are correct.

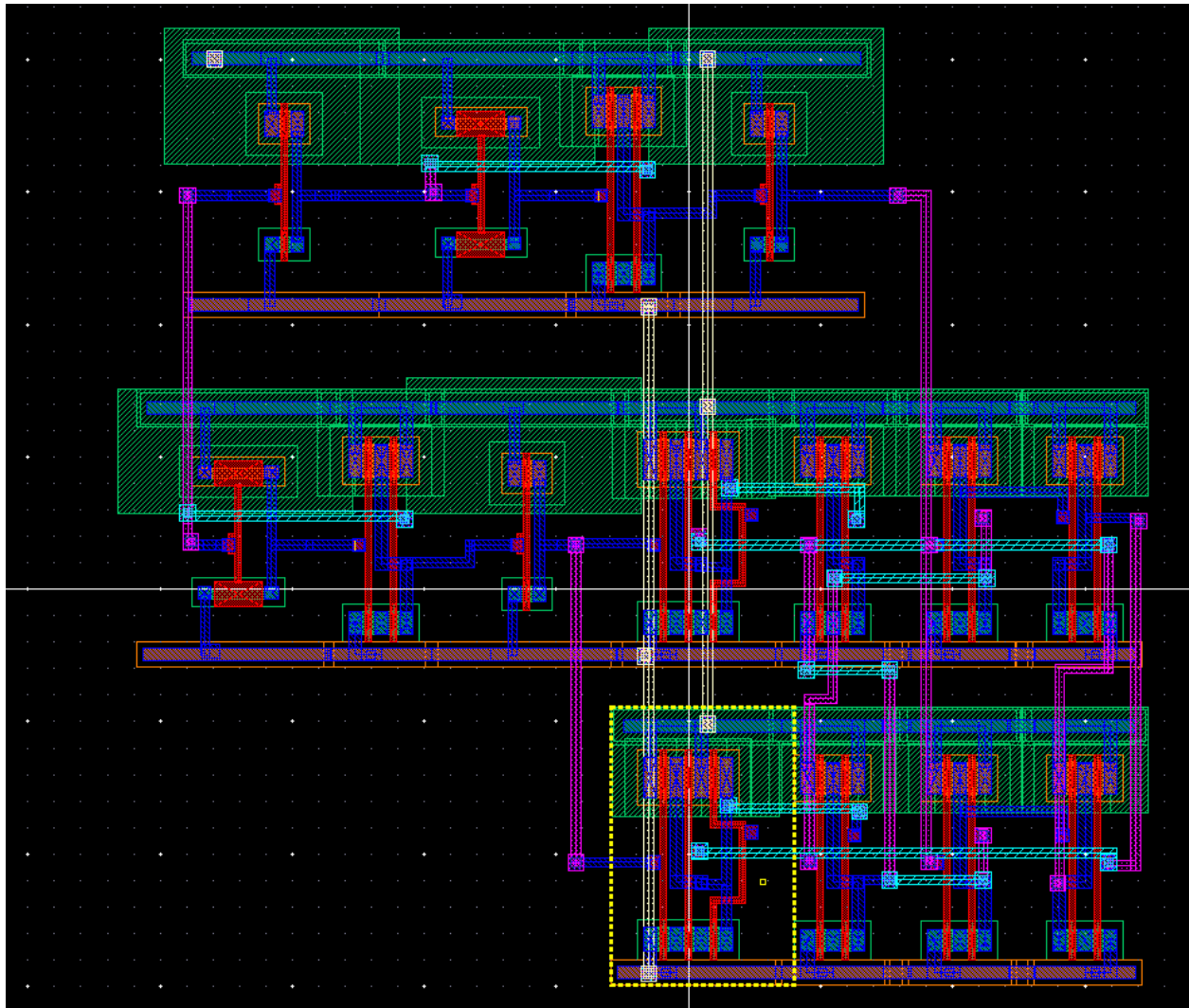
Theory

The binary counter that is built from a sequential digital logic will store 1 bit and update it synchronously with the clock, when JK FF makes its output to 1, it will cascade it to make a binary counting sequence.

JK Flip-Flop schematic



JK Flip-Flop layout

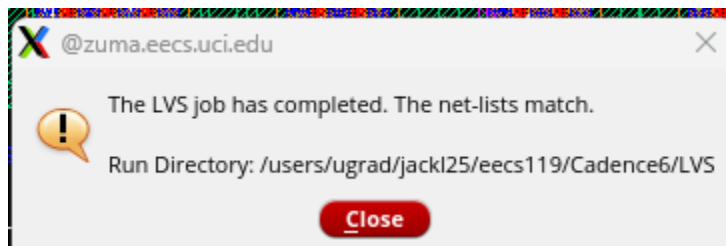
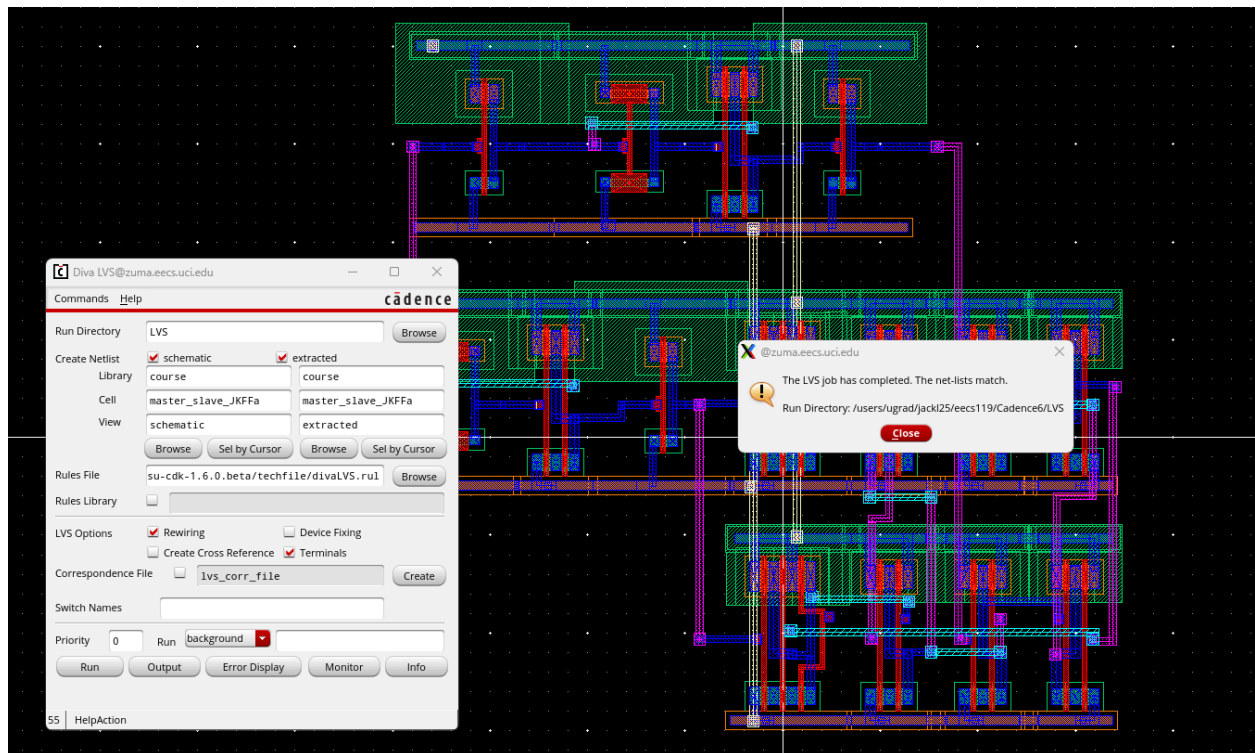


JK Flip-Flop DRC

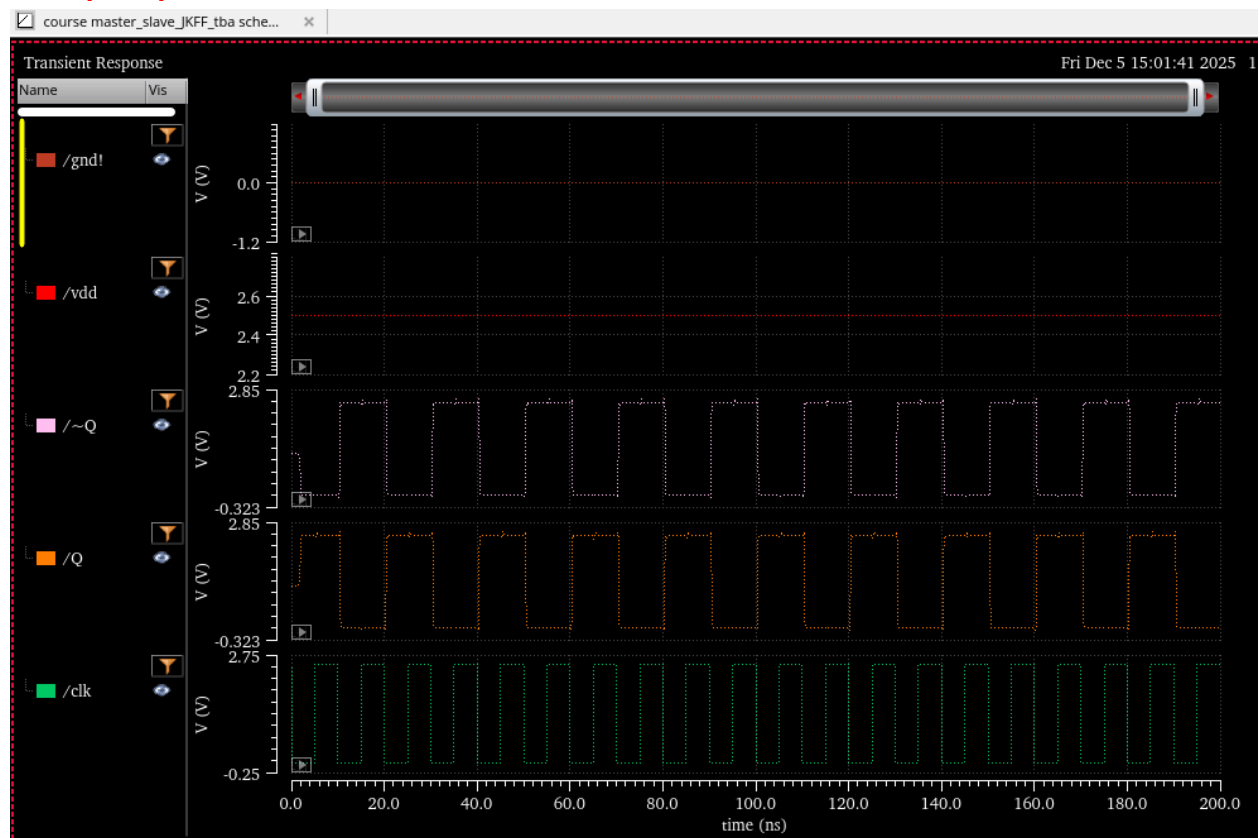
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Total errors found: 0
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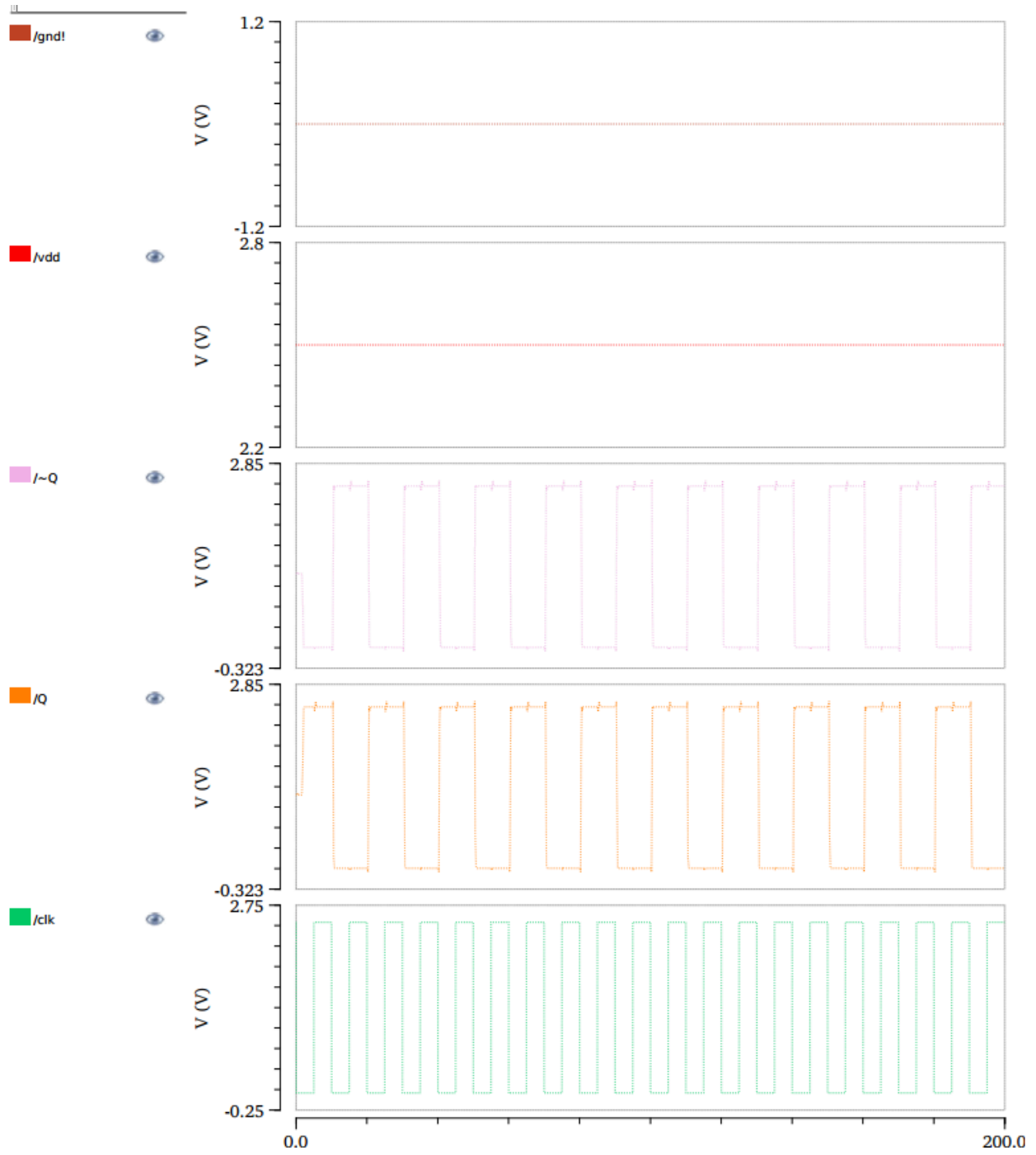
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Extraction started at Fri Dec 5 17:17:13 2025
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JK Flip-Flop LVS

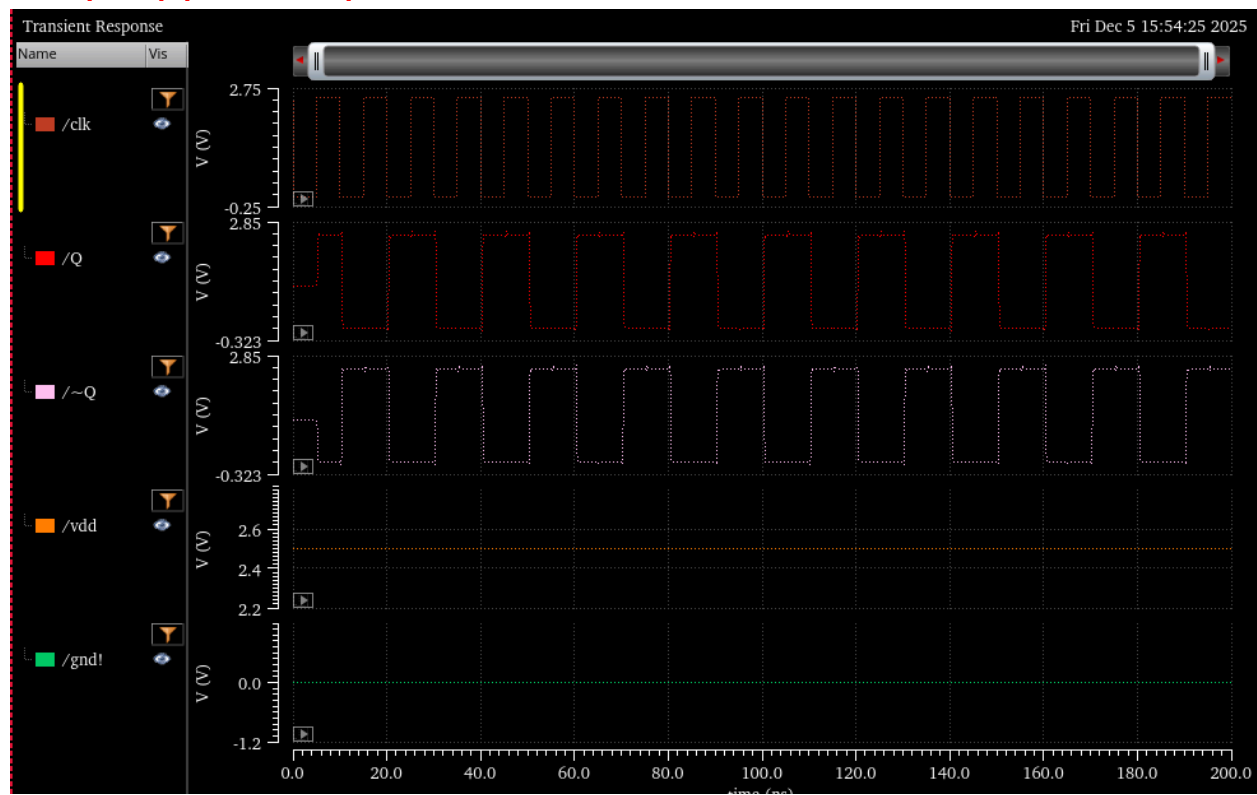


JK Flip-Flop Simulation





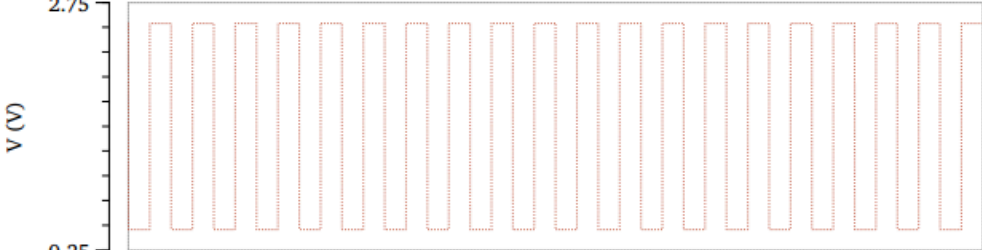
JK Flip-Flop parasitic capacitance simulation



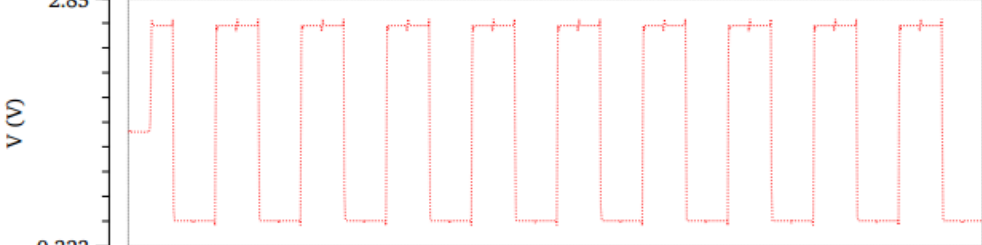
Transient Response

Fri Dec 5 15:54:25 2025

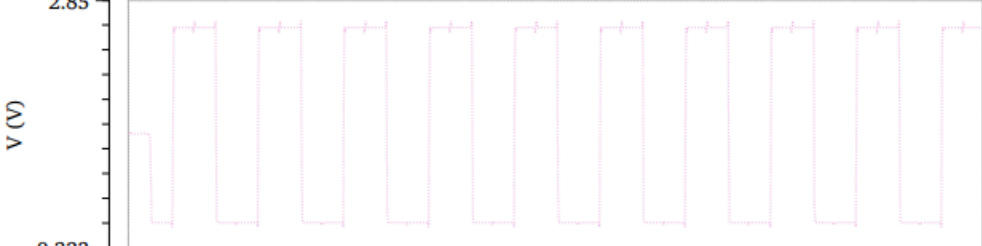
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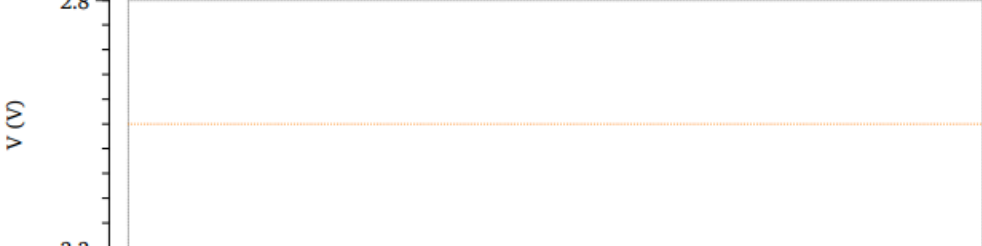
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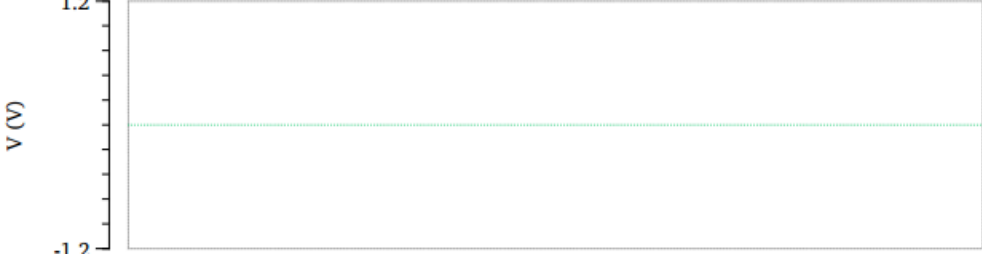
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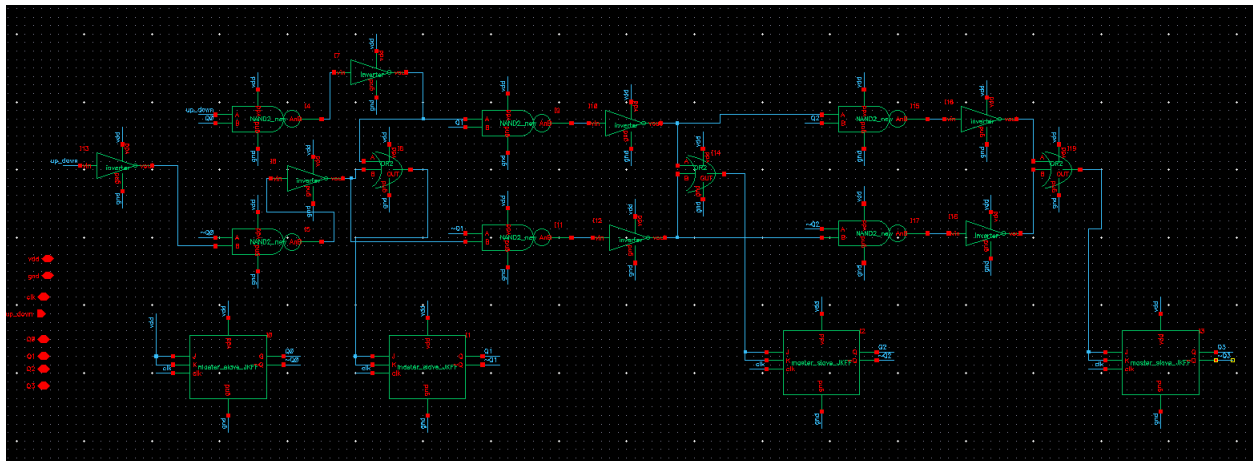
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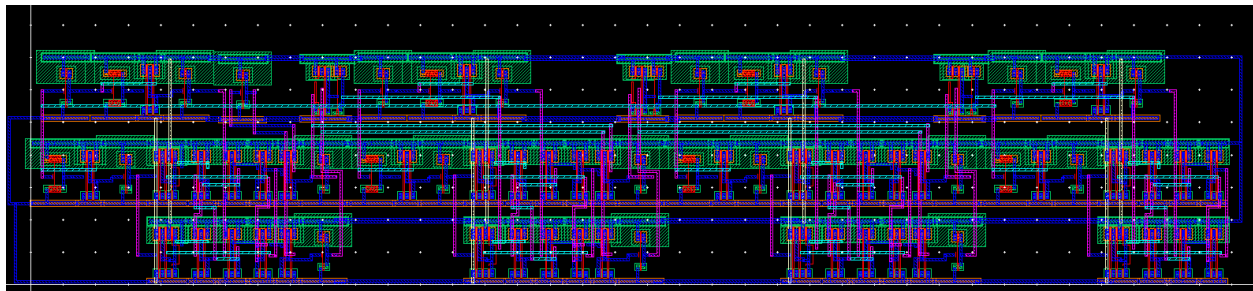
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4 Bit Up-Down schematic



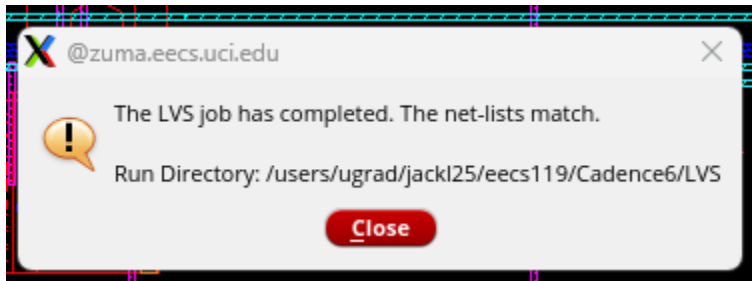
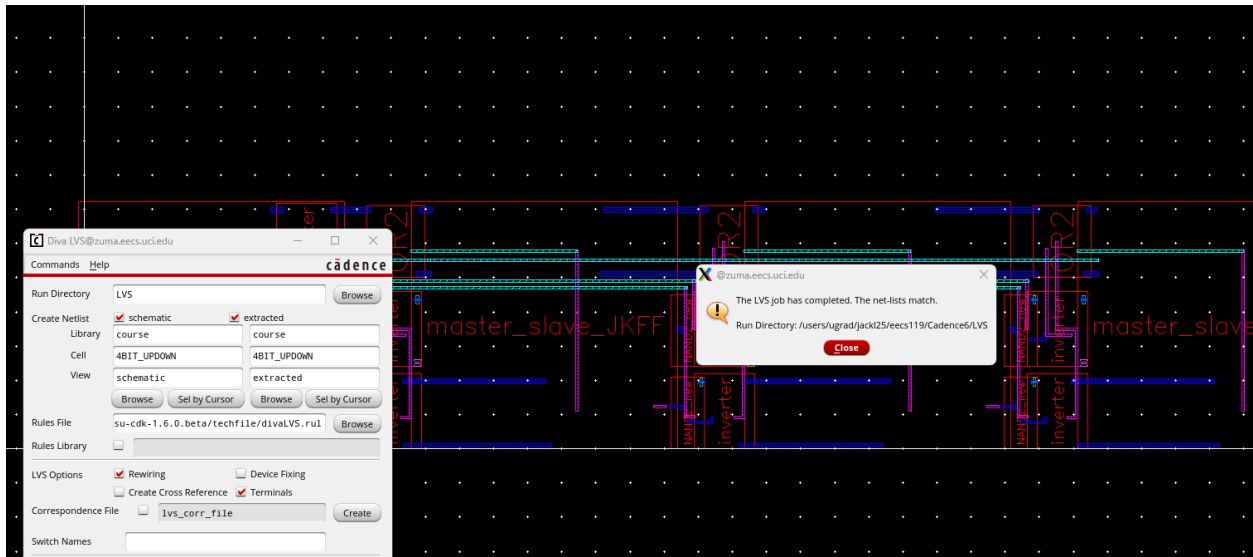
4 Bit Up-Down layout



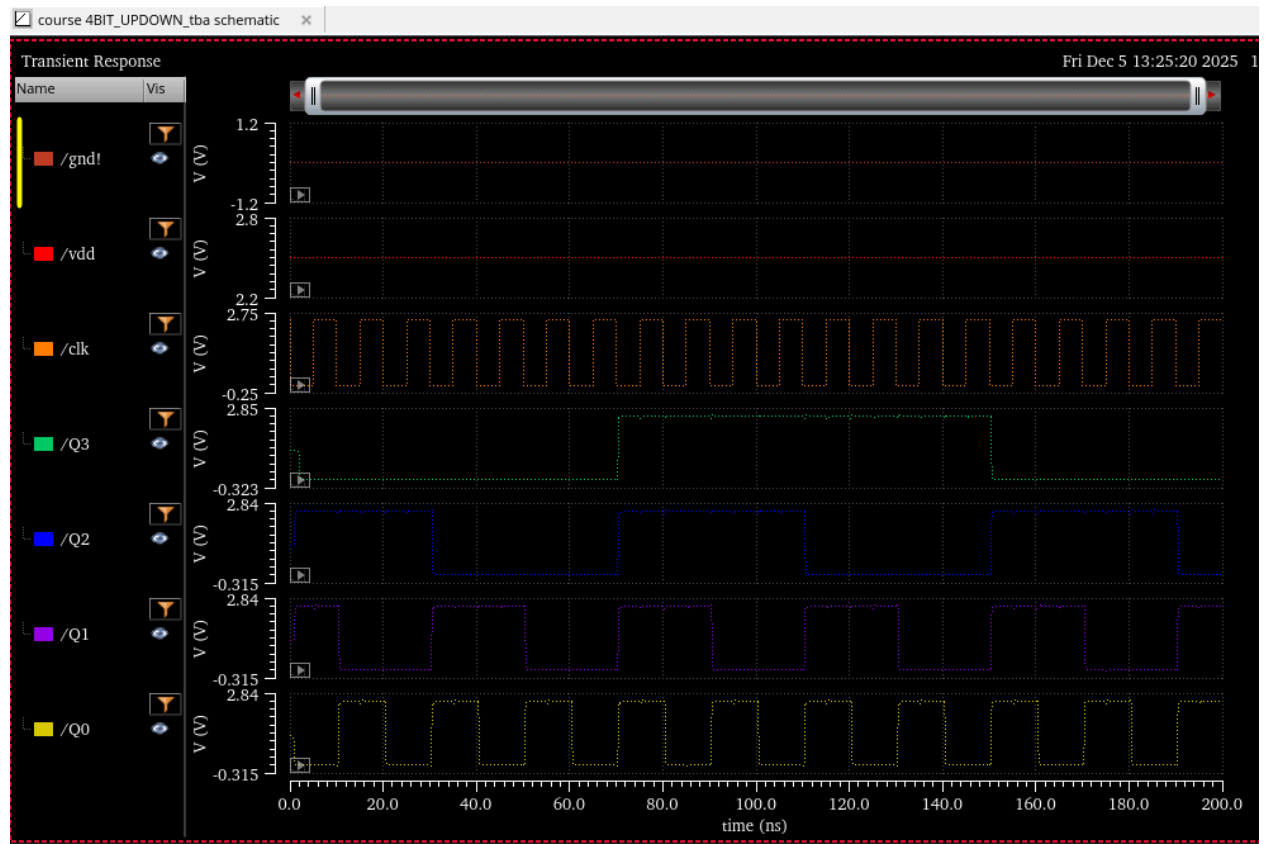
4 Bit Up-Down DRC

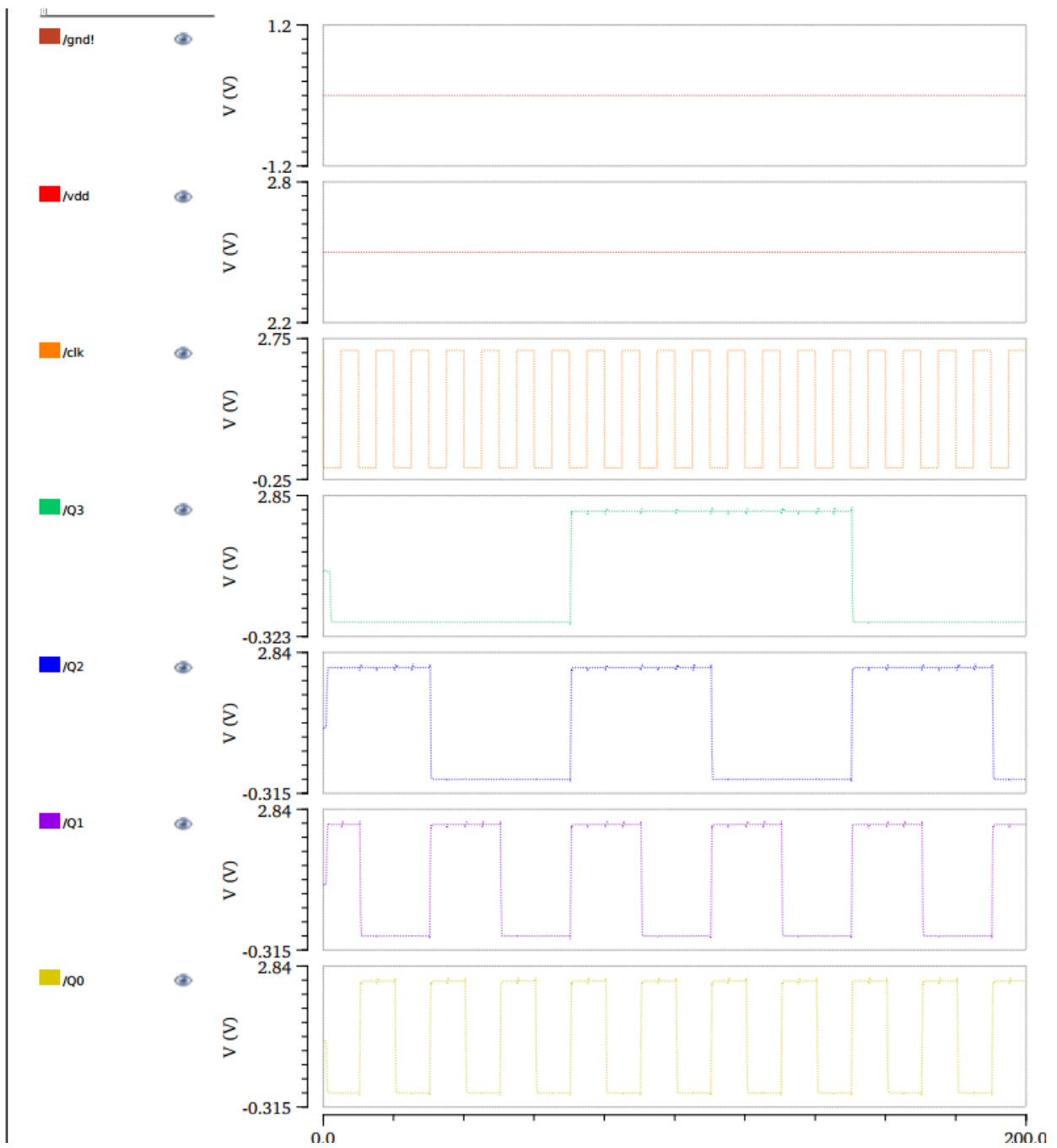
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***** Summary of rule violations for cell "4BIT_UPDOWN layout" *****
Total errors found: 0
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4 Bit Up-Down LVS

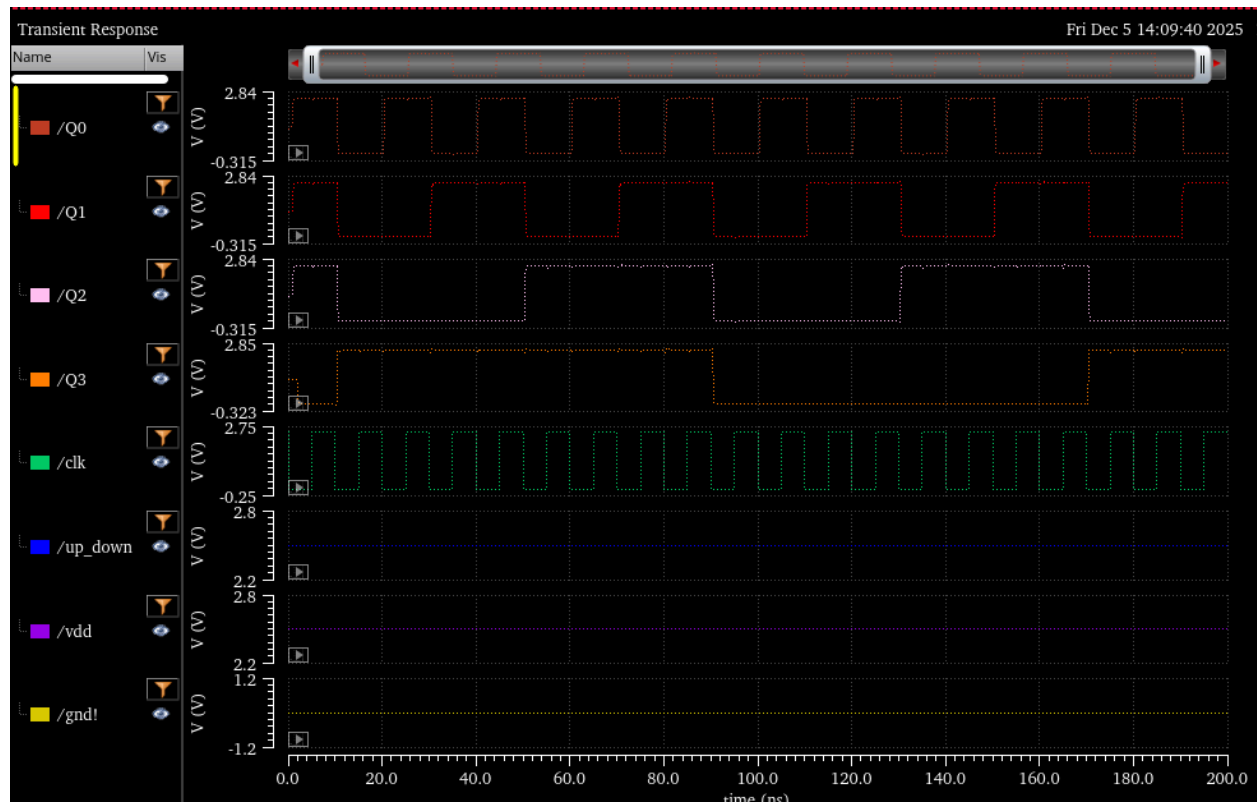


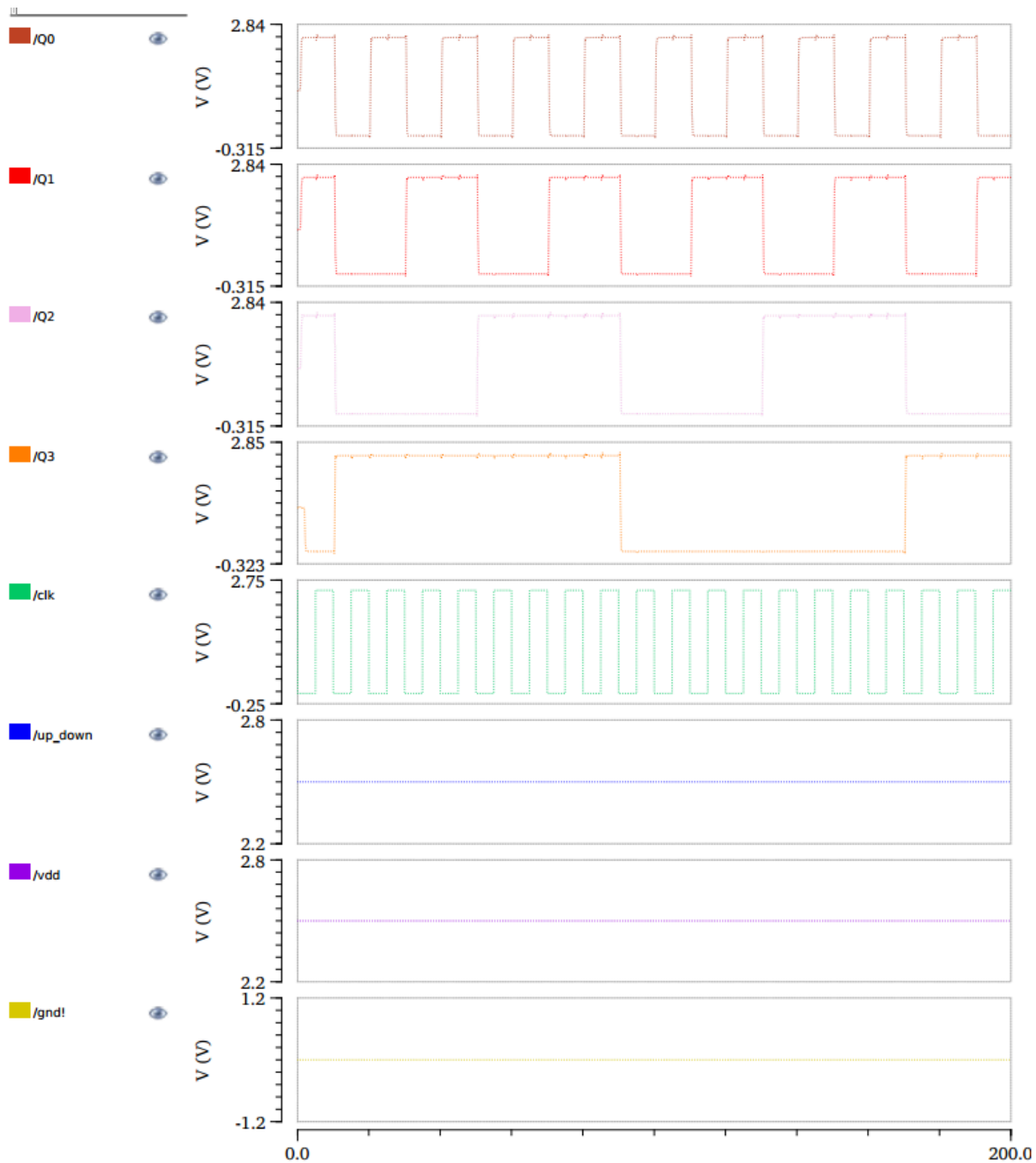
4 Bit Up-Down - down



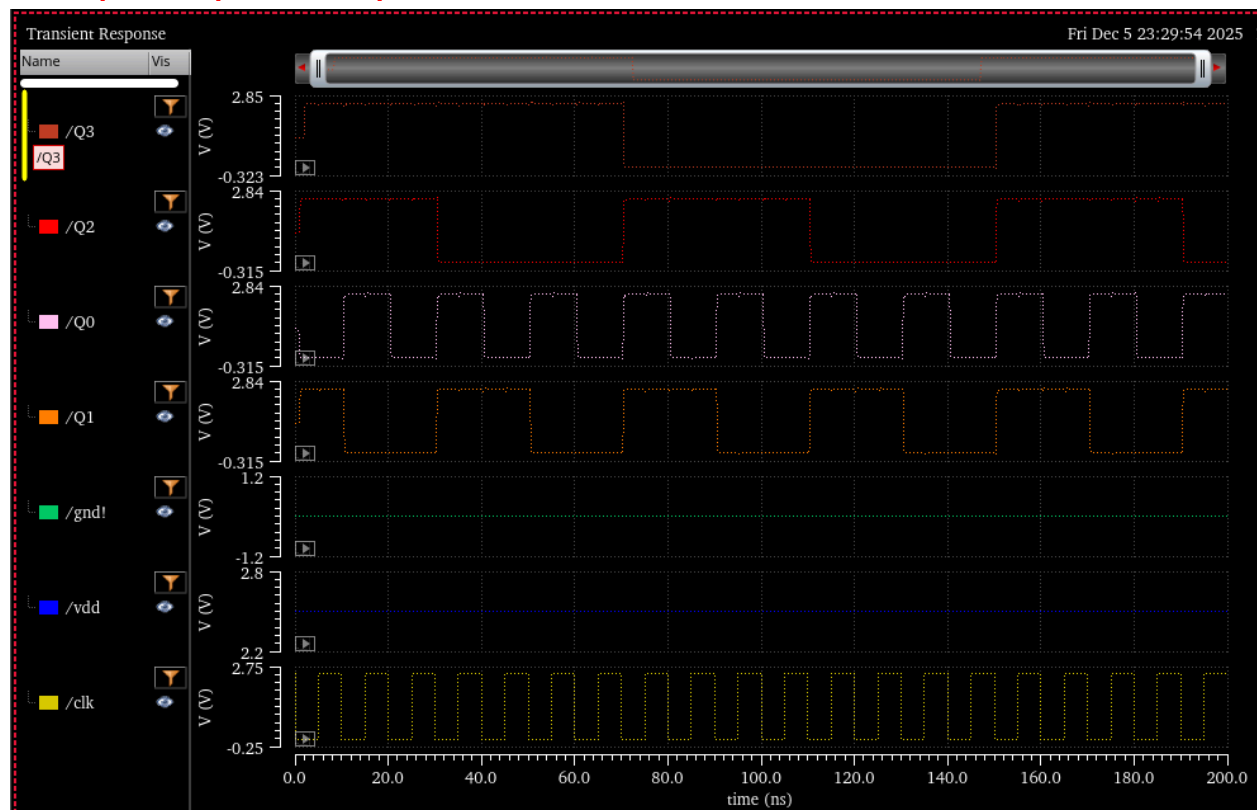


4 Bit Up-Down - up



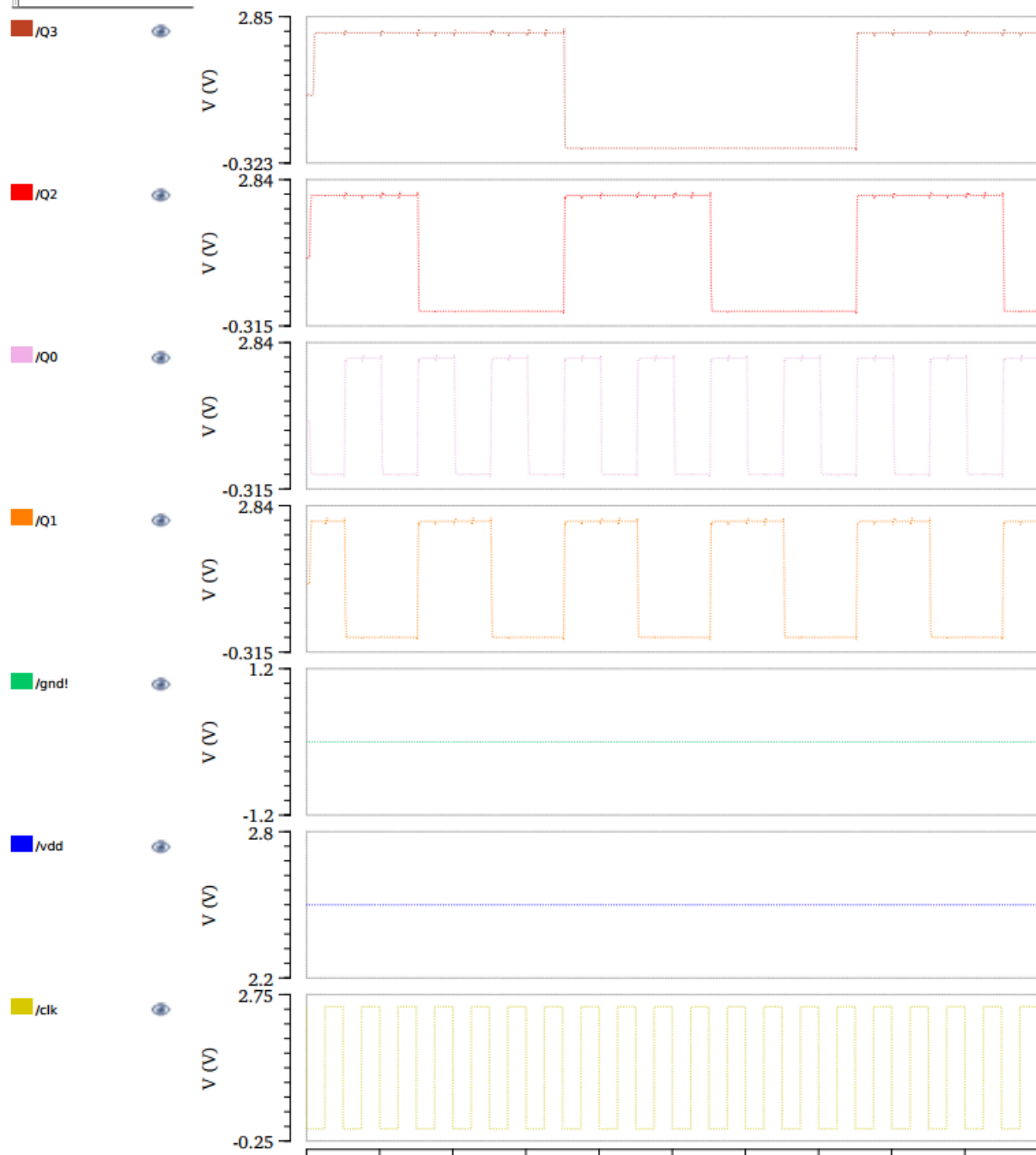


4 Bit Up-Down parasitic capacitance simulation - down

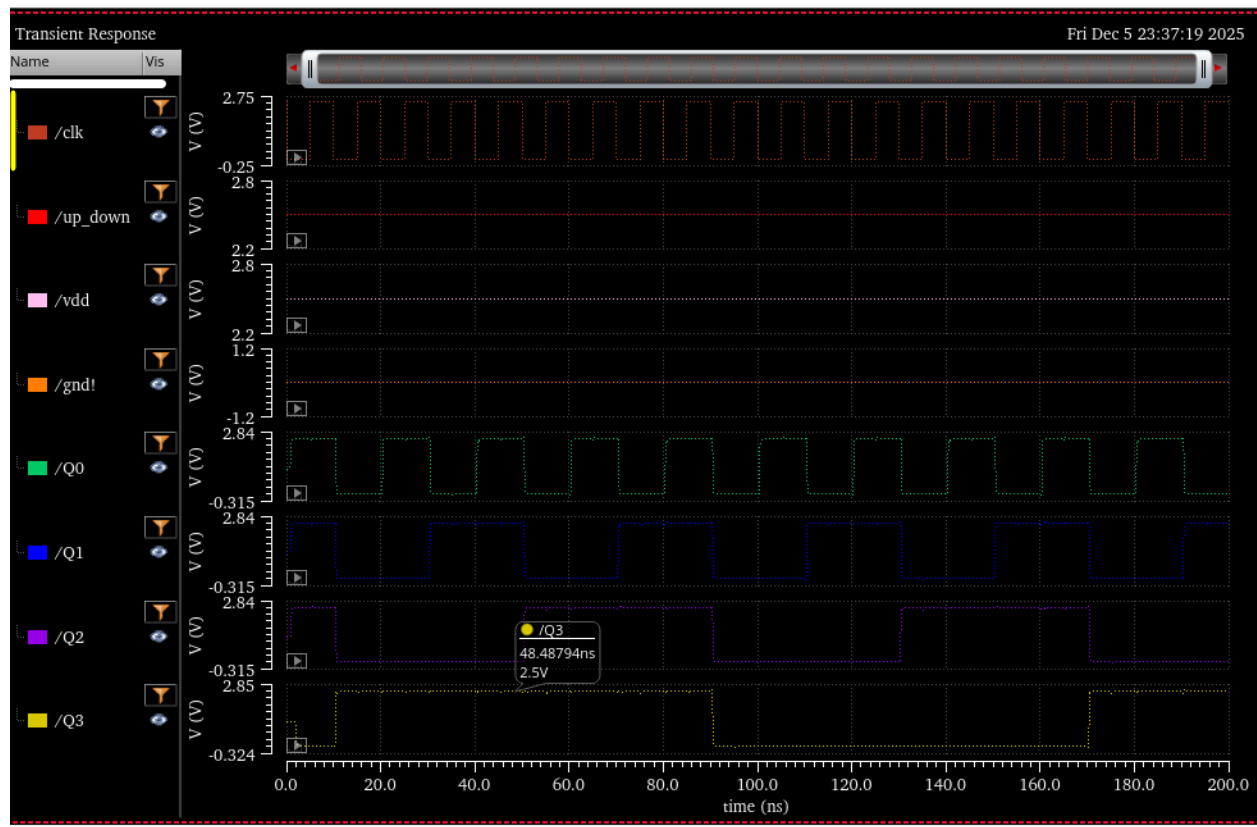


Transient Response

Fri Dec 5 23:29:54 2025

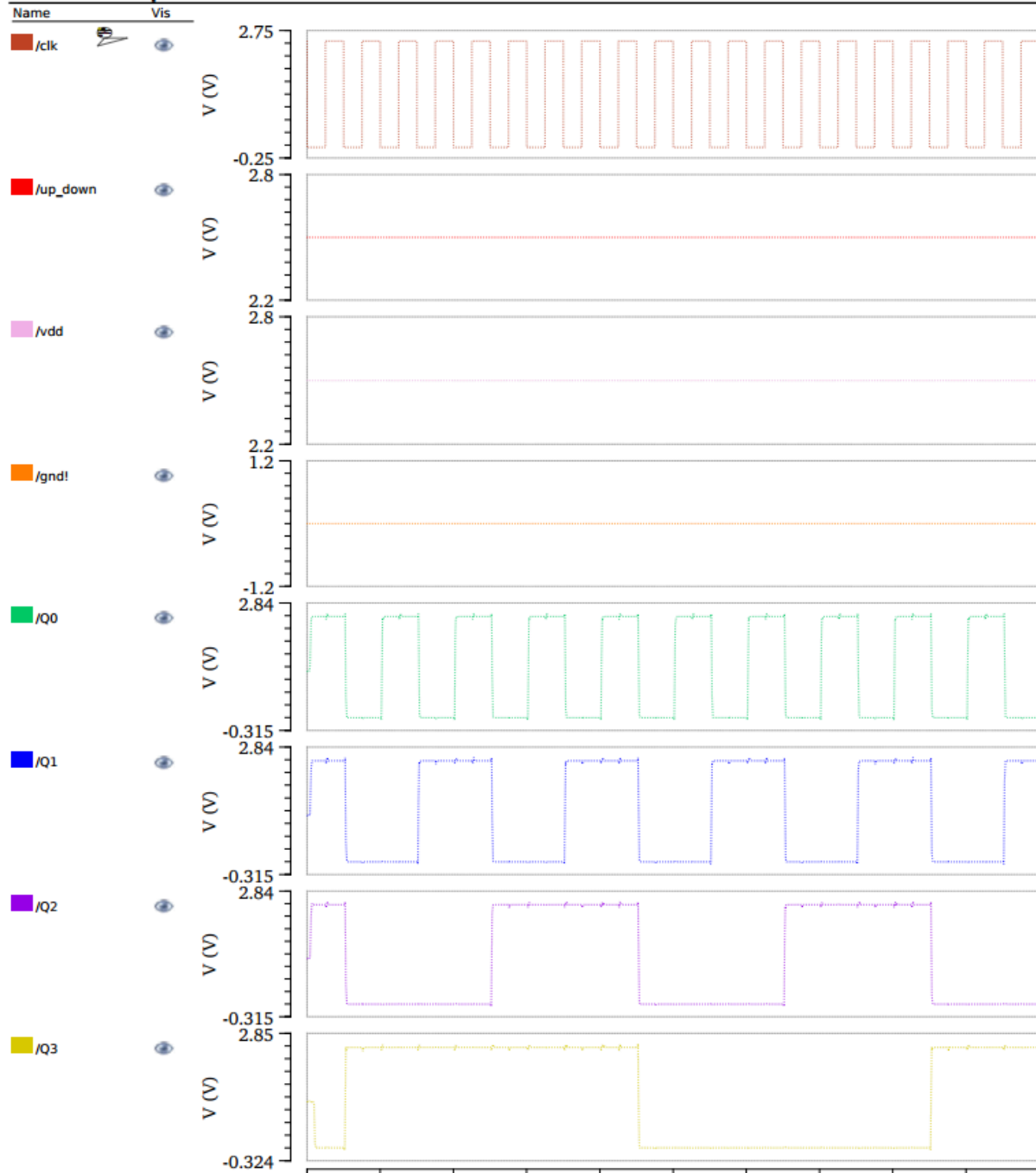


4 Bit Up-Down parasitic capacitance simulation - up



Transient Response

Fri Dec 5 23:37:19 2025



Conclusion

The simulation of both the JK FF and 4Bit up-down was successful with both of them passing the LVS and DRC check, the result from the simulation verifying the increasing values and down values working correctly depending on if it was 0v or 2.5v.